4) UDP scan - Send UDP packets & can identify an active host through an ICMP port

OBSERVATION:

→ ARP request can discover hosts within the same subnet

→ ARP cannot cross routers because it operates at the link layer.

→ An ARP ping generally requires a previledged account.

→ UDP ping can identify an active host when an ICMP port unreachable is received.

→ The -sn option allows Nmap to perform host discovery without conducting a port scan.

RESULT:

Thus, the live host discovery using Nmap was successfully studied and performed. The experiment helped in understanding how active hosts can be identified before performing further network scans.

24/9/25

EXP. NO: 8
DATE: 17/9/26

DISCOVERY OF LIVE HOST USING ARP, ICMP, TCP, UDP

AIM:

To discover live host in a network using Nmap scans such as ARP, ICMP, TCP and UDP on the TryHackMe platform.

OBJECTIVE:

To understand how nmap identifies ~~hosts~~ active systems before performing further network scanning.

SOFTWARE:

Nmap
Tryhack Nmap loom
Computer with network success

THEORY:

Nmap is a free and open-source network scanning tool used to discover live hosts and identify network services.
It was created

Nmap uses different live hosts using different protocols.

1) ARP scan - Uses ARP scans request to find hosts on same subnet.
2) ICMP scan - Use ICMP request and replies to identify active host.
3) ~~Use~~ TCP scan - Use TCP requests and packets to specified ports & check the response.